

Question	Answer	Marks
4(a)(i)	time for one oscillation/one vibration/one cycle or time between adjacent wavefronts/points in phase or shortest time between two wavefronts/points in phase	B1
4(a)(ii)	distance moved by wavefront/energy during one cycle/oscillation/period (of source) or minimum distance between two wavefronts or distance between two adjacent wavefronts or minimum distance between two points having the same displacement and moving in the same direction	B1
4(b)(i)	$v = \lambda / T$ or $v = f\lambda$ <u>and</u> $f = 1 / T$	C1
	$\lambda = 20 \times 0.60$ $= 12 \text{ cm}$	A1
4(b)(ii)	phase difference $= 360^\circ \times (0.20/0.60)$ or $360^\circ \times (0.40/0.60)$ $= 120^\circ$ or 240°	A1
4(b)(iii)	$I \propto A^2$	C1
	$I_Q / I_P = A_Q^2 / A_P^2$ $= 2.0^2 / 3.0^2$ $= 0.44$	A1
4(b)(iv)	displacement $= 1.00 - 3.00$ $= -2.00 \text{ mm}$	A1

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5(c)(i)	waves spread out (each) slit more	B1